

Answer **all** questions in the spaces provided.

1. Drinks manufacturers use standard food colourings which are identified by E-numbers. A student was asked to investigate which E-numbers were present in different brands of orange drinks.

The student suggested the following method:

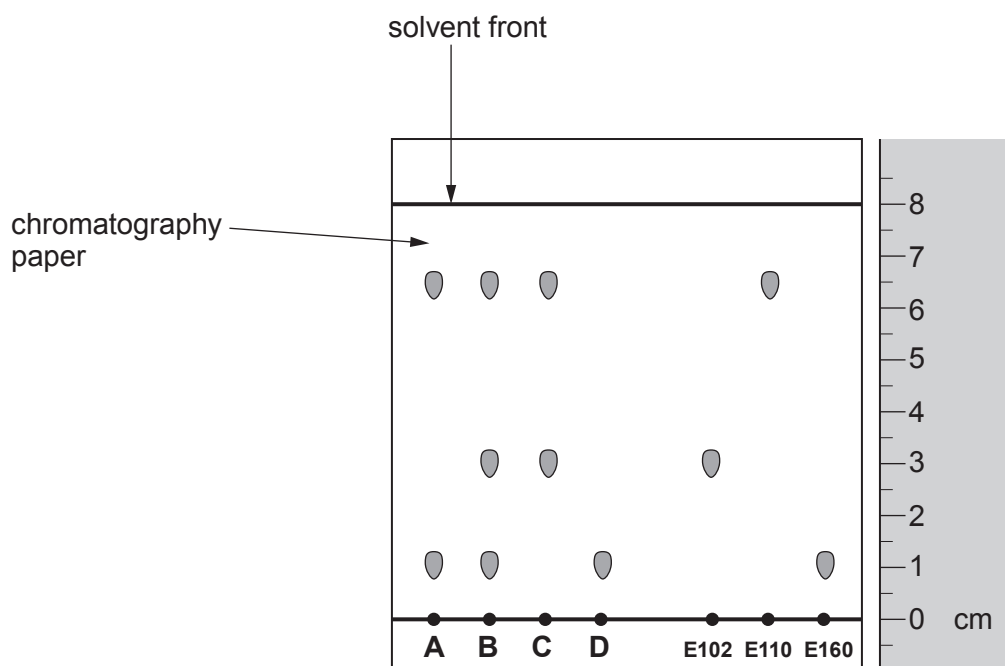
1. Place spots of each drink **A**, **B**, **C** and **D** on a pen line on chromatography paper.
2. Place spots of three common food colourings, **E102**, **E110**, **E160** on the pen line.
3. Hang the chromatography paper in a beaker and add water until it is above the pen line.
4. Leave the paper to stand in water in a beaker until the water rises to near the top of the chromatography paper.

- (a) The student failed to produce a chromatogram using this method. State **two** errors in this method. [2]

1.

2.

- (b) Another student, using a valid method, obtained the chromatogram below (*not drawn to scale*).



Answer the following questions using the information in the chromatogram.

- (i) State which food colouring (**E102**, **E110**, **E160**) is the most soluble. [1]

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- (ii) State which drinks (**A**, **B**, **C** or **D**) contained **E110**. [1]

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- (iii) The student predicted that all orange drinks contain the same E-numbers. Explain whether this prediction was correct by comparing the food colourings found in drinks **B** and **D**. [2]

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- (iv) The retention factor (R_f) value of a substance can be used to identify that substance from a chromatogram.

- I. State the distance travelled by the solvent front. [1]

Distance = cm

- II. Use the equation:

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by the solvent front}}$$

- to calculate the R_f value for **E102**. [2]

R_f =